

Farmer's Market Management System Project Report

**Software Engineering Concepts
EEX4347**

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Farmer's Market Management System

A local farmer's market operates weekly, bringing together farmers and local producers to sell fresh produce and handmade goods. Currently, vendor registration, stall reservations, event scheduling, and product listings are managed manually, often resulting in confusion, overbooking, and miscommunication. To address these issues, the market organizers propose the implementation of an Online Farmer's Market Management System. This system will enable farmers to register as vendors, update their profiles, and reserve stalls for upcoming market days. They can also list their products, such as fruits, vegetables, dairy, and handmade items, along with descriptions, prices, and availability.

Customers will be able to view the listed products, make reservations for pickup on market day, and communicate with vendors through a built-in messaging system to confirm orders or ask questions. The system also allows market administrators to manage vendor registrations, monitor stall assignments, schedule market events, and generate reports on vendor participation and product reservations.

✦ Introduction

The Farmer's Market Management System is a simple web-based application developed to connect customers with local vendors. Customers can browse available products, make reservations, and vendors can manage their products easily. The system also allows events to be announced to users.

This project uses a PHP-based web front-end with a MySQL database (WAMP server). The main purpose is to understand basic system development, database design, and user authentication.

✦ Problem Statement

Traditional farmers' markets often struggle with organizing product availability, vendor registration, and customer reservations. There is no online system to manage products, events, and customer needs. Therefore, a simple centralized system is required to:

- Maintain vendor details
- Display available products
- Allow customers to reserve items
- Manage events
- Provide secure login for users

✦ Objectives

Main Objectives

- To develop a simple web system for product viewing and reservation.
- To create a structured MySQL database with tables for users, vendors, customers, products, reservations, and events.
- To implement login and signup for user authentication.

Additional Objectives

- To practice connecting PHP pages to a MySQL database.
- To organize and store market information efficiently.
- To demonstrate CRUD operations (Create, Read, Update).

★ System Overview

The system consists of three main user roles:

Customer

- Signup/Login
- View products
- Make reservations
- View events

Vendor

- Signup/Login
- Add products
- Update availability
- View customer reservations

Admin

- Manage events
- View all users and vendors

★ Functional Requirements

1. User Login & Signup

- Users can register with email, password, and role.
- Login validation for all users.

2. Vendor Functions

- Add product details (name, category, price, availability).
- Manage product availability.

3. Customer Functions

- Browse products.
- Make reservations.
- View events.

4. Event Management

- Admin can publish events (date + description).

5. Database Management

- Store users, vendors, customers, products, reservations, and events.

★ Non-Functional Requirements

- **Usability:** Simple and easy-to-understand interface.
- **Performance:** Fast loading pages due to lightweight PHP and MySQL.
- **Reliability:** Stable database connections using db.php.
- **Security:** Password protection (basic level).

★ Database Design

Below is the final database structure used for this project.

Customer Table

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	customer_id	 int			No	None			 Change  Drop  More
☐	2	name	varchar(100)	utf8mb4_unicode_ci		No	None			 Change  Drop  More
☐	3	email	varchar(100)	utf8mb4_unicode_ci		No	None			 Change  Drop  More
☐	4	phone	varchar(20)	utf8mb4_unicode_ci		No	None			 Change  Drop  More

Vendor Table

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	vendor_id	 int			No	None			 Change  Drop  More
☐	2	name	varchar(100)	utf8mb4_unicode_ci		No	None			 Change  Drop  More
☐	3	email	varchar(100)	utf8mb4_unicode_ci		No	None			 Change  Drop  More
☐	4	phone	varchar(20)	utf8mb4_unicode_ci		No	None			 Change  Drop  More
☐	5	farm_details	text	utf8mb4_unicode_ci		No	None			 Change  Drop  More
☐	6	status	enum('Pending', 'Approved', 'Rejected', '')	utf8mb4_unicode_ci		No	None			 Change  Drop  More

Product Table

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	product_id	int			No	None			 Change  Drop  More
☐	2	vendor_id	 int			No	None			 Change  Drop  More
☐	3	name	varchar(100)	utf8mb4_unicode_ci		No	None			 Change  Drop  More
☐	4	category	varchar(50)	utf8mb4_unicode_ci		No	None			 Change  Drop  More
☐	5	price	decimal(10,2)			No	None			 Change  Drop  More
☐	6	availability	int			No	None			 Change  Drop  More

Reservation Table

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	reservation_id	int			No	None			Change Drop More
☐	2	customer_id	int			No	None			Change Drop More
☐	3	product_id	int			No	None			Change Drop More
☐	4	quantity	int			No	None			Change Drop More
☐	5	status	enum('Pending', 'Confirmed', 'Cancelled', '')	utf8mb4_unicode_ci		No	None			Change Drop More

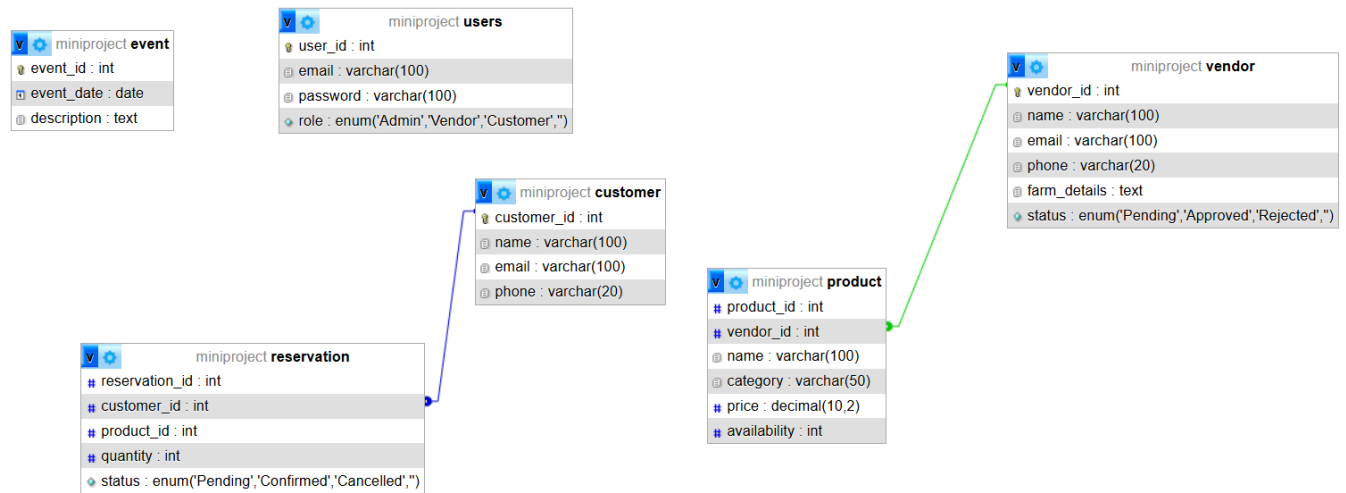
Event Table

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	event_id	int			No	None			Change Drop More
☐	2	event_date	date			No	None			Change Drop More
☐	3	description	text	utf8mb4_unicode_ci		No	None			Change Drop More

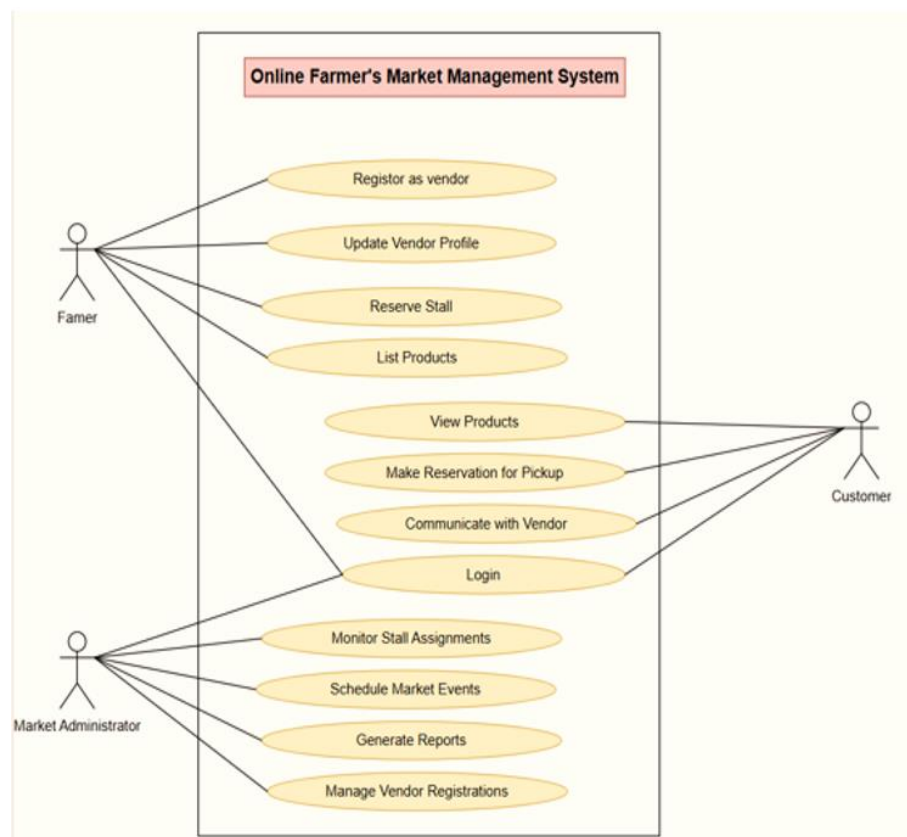
user Table

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	user_id	int			No	None		AUTO_INCREMENT	Change Drop More
☐	2	email	varchar(100)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	3	password	varchar(100)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	4	role	enum('Admin', 'Vendor', 'Customer', '')	utf8mb4_unicode_ci		No	None			Change Drop More

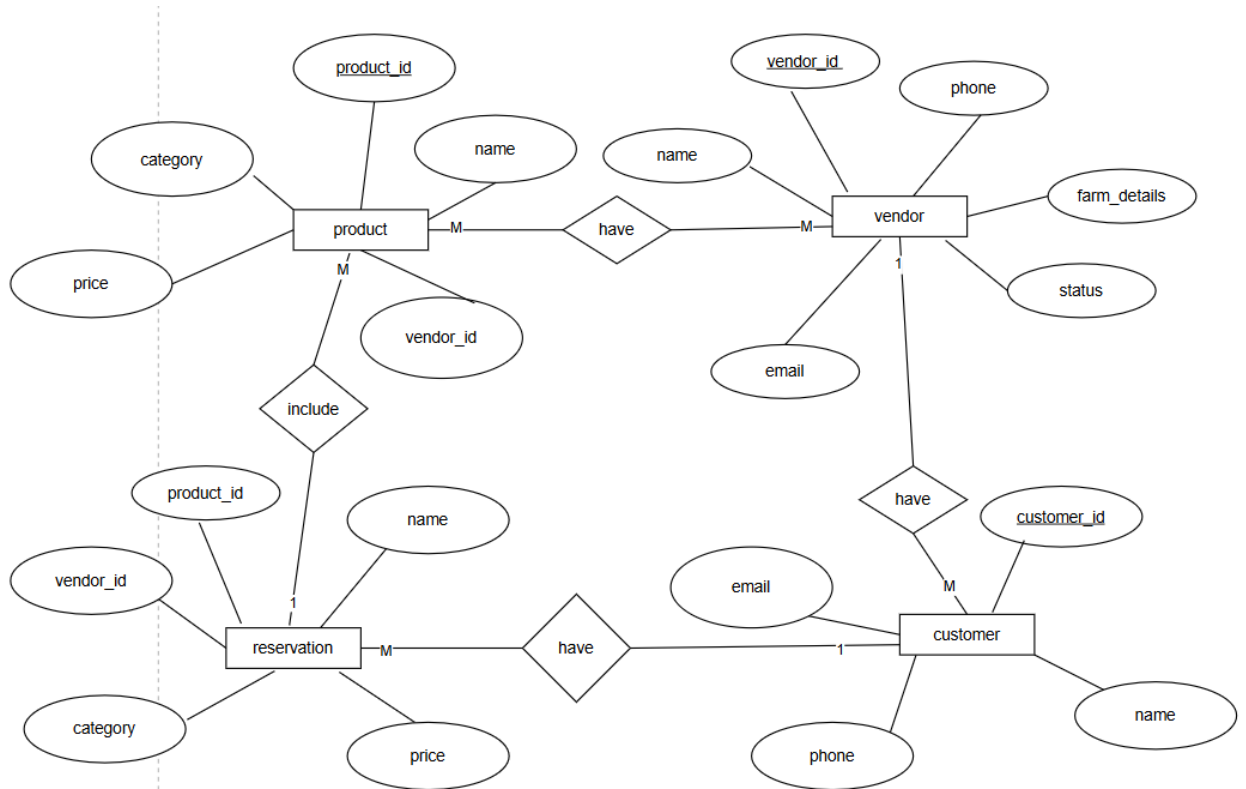
Database schema



Diagrams



ER diagram



✦ System Implementation

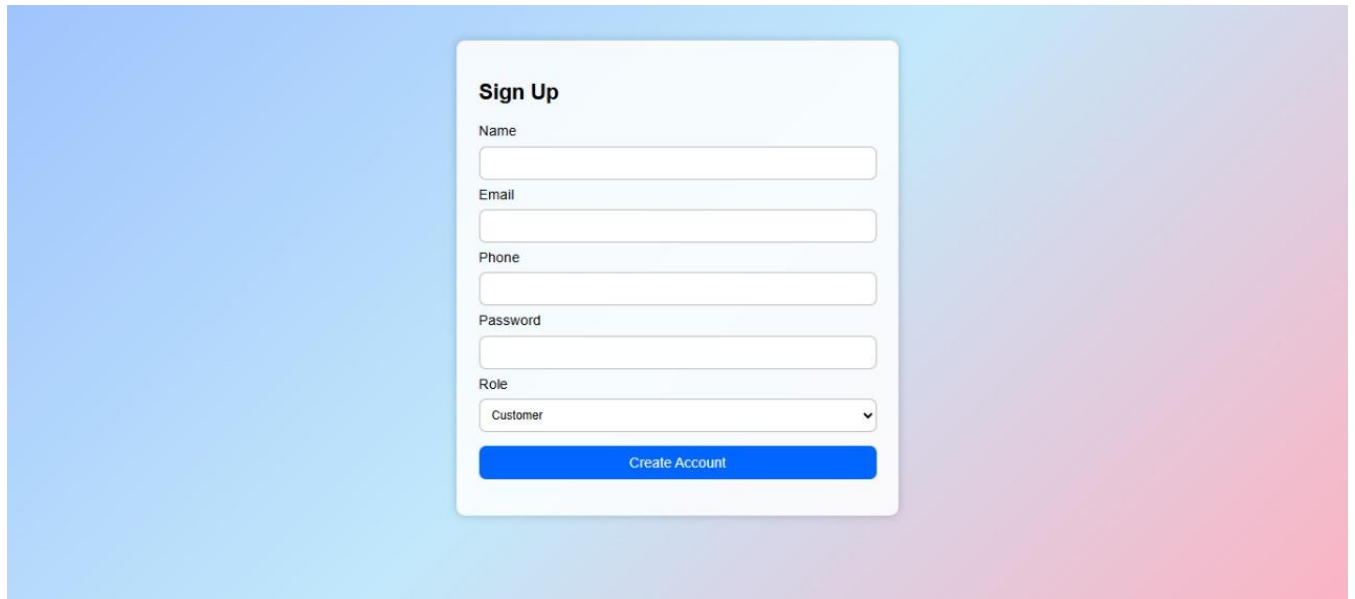
Technologies Used

- **Frontend:** HTML, CSS
- **Backend:** PHP
- **Database:** MySQL
- **Local Server:** WAMP

✦ Key Pages Implemented

1. **login.php** – allows users to log into the system
2. **signup.php** – creates new user accounts
3. **vendor_add_product.php** – vendors add products
4. **browse_products.php** – customers view products
5. **reserve.php** – customers make reservations

★ Sample Screenshots



A screenshot of a 'Sign Up' form centered on a background with a blue-to-pink gradient. The form is a white card with a light blue header. It contains input fields for Name, Email, Phone, and Password, followed by a Role dropdown menu set to 'Customer'. A blue 'Create Account' button is at the bottom.

Sign Up

Name

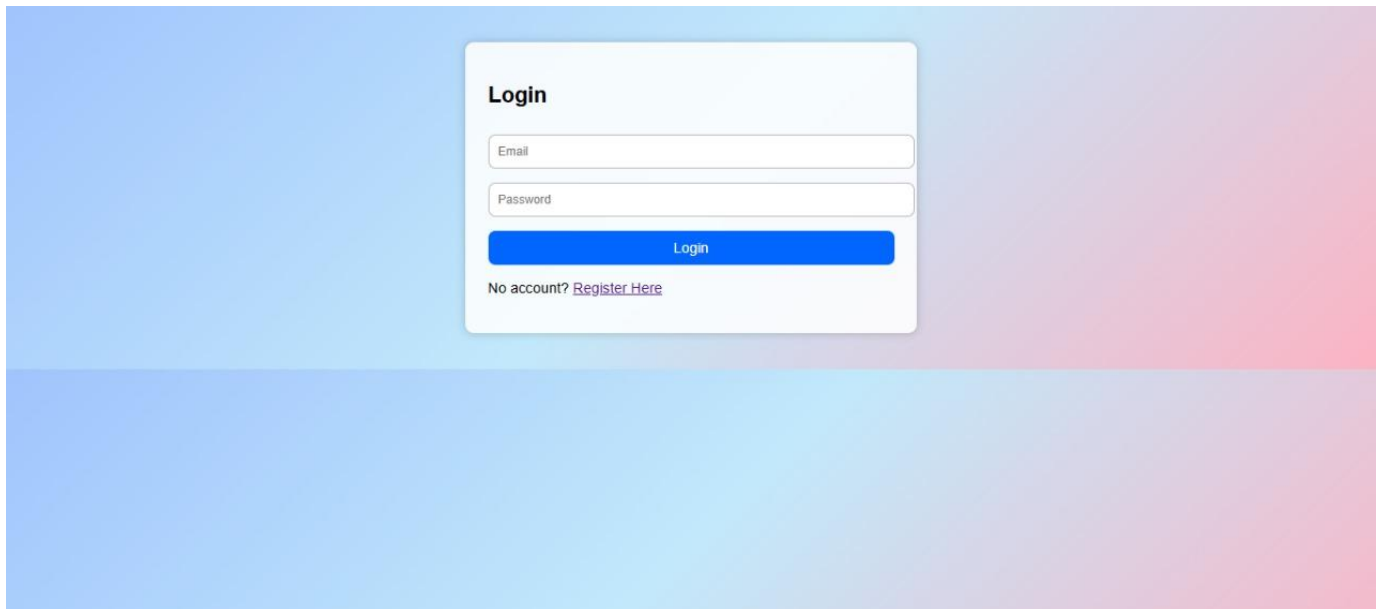
Email

Phone

Password

Role

[Create Account](#)



A screenshot of a 'Login' form centered on a background with a blue-to-pink gradient. The form is a white card with a light blue header. It contains input fields for Email and Password, followed by a blue 'Login' button. Below the button is a link for users without an account.

Login

Email

Password

[Login](#)

No account? [Register Here](#)

Available Products

[Logout](#)

Welcome! You are logged in as: **Customer**

Product	Category	Price	Stock	Reserve
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★ Testing

Test Case	Input	Expected Output	Result
Login with correct email + password	Valid data	Redirect to role page	Pass
Login with wrong password	Wrong password	Error message shown	Pass
Signup new user	Email, password, role	Saved in database	Pass
Vendor adds product	Product details	Stored in product table	Pass
Customer reservation	Quantity selected	Entry in reservation table	Pass

★ Problems Faced & Solutions

1. **db.php not found error**

Cause: db.php was missing

Solution: Created db.php and placed inside project folder.

2. **Cannot access pages directly**

Cause: Opening files through VS Code

Solution: Used URL <http://localhost/EEX4347/>.

3. **Foreign key errors**

Cause: Missing index on vendor_id

Solution: Created primary key and index.

✦ Conclusion

This mini project successfully demonstrates a simple but functional Farmer's Market Management System using PHP and MySQL. The database is well-structured with six interconnected tables, enabling customers, vendors, and events to be managed in an organized way.

This project helped develop practical skills in system design, database management, and basic web development.

✦ Future Improvements

- Add payment functionality
- Add profile pages for customers and vendors
- Add admin dashboard
- Improve UI using CSS frameworks
- Add real-time stock updates